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Description - This procedure relates to Signa Horizon products having GRAM 1 modules. If you have SGD miniGRAM's refer to the procedure for DC Offset Adjustment - MiniGRAM.

Before shimming, perform the GRAM DC Offset Adjustment on each axis of the GRAM. Failure to do so will cause errors in shimming.

1- PRELIMINARY SET-UP

1. Remove the front cover of the GRAM Cabinet.
2. Remove the cover on each GRAM module for each axis being calibrated.

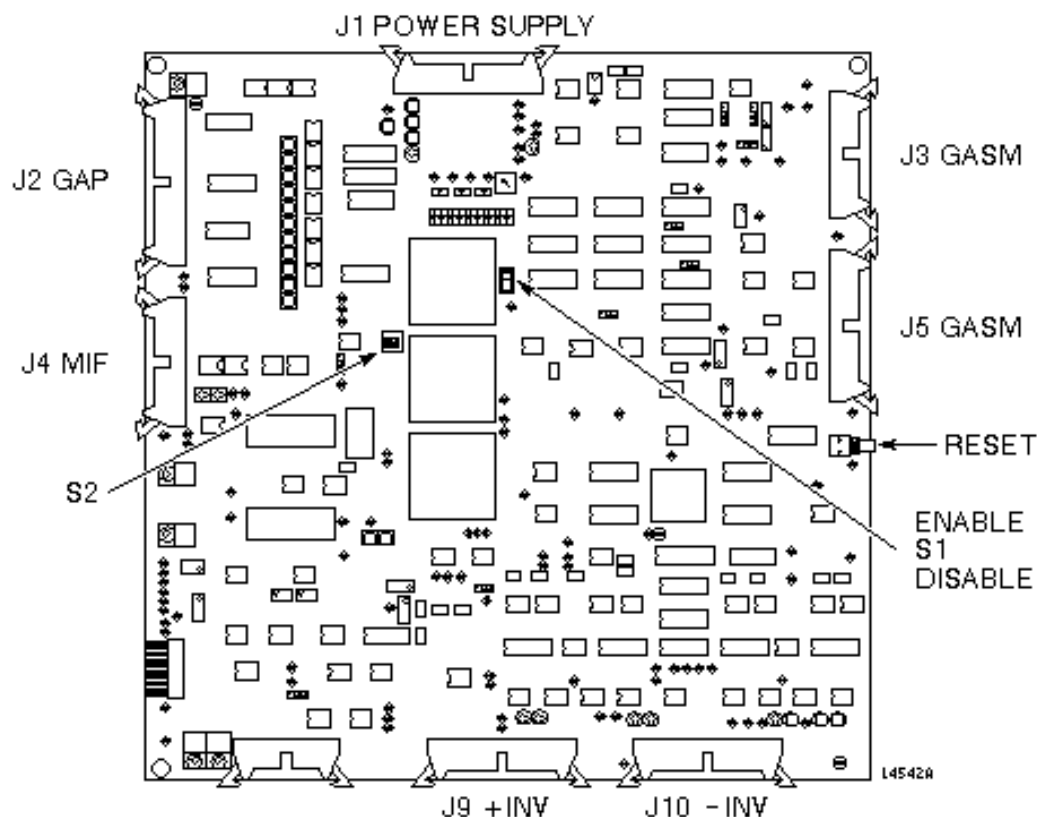
Note

Verify switch & jumper settings - Verify that all switches and jumpers on the GRAM Control Board and the Analog Tuning Board are set to the proper position. Failure to do so will cause adjustments to be inaccurate.

3. Verify the position of all jumpers and switches on the GRAM Control Board and the GRAM Power Supply Board for each axis.
 - a. See Table 1 and Illustration L4542A for proper switch settings for the GRAM Control Board.

TABLE 1
GRAM CONTROL BOARD SWITCH SETTINGS

Number	Position	Setting	Name & Position Definition
S1	N/A	Up	GRAM Enabled: When S1 is down, the GRAM is disabled.
S2	1	Down	(RES) Reserved: unused
S2	2	Down	(HV) High Voltage Enable: Under software control
S2	3	Down	(FW) Free Wheel: Enables PWM mode under software control. Up is Freewheel mode, no PWM.
S2	4	Down	(MAN) Manual Ready: Manual Ready is controlled by GAP. Up is Troubleshooting; it forces the GRAM to Manual Ready. (S1 must also be up or enabled when MAN is Up.)

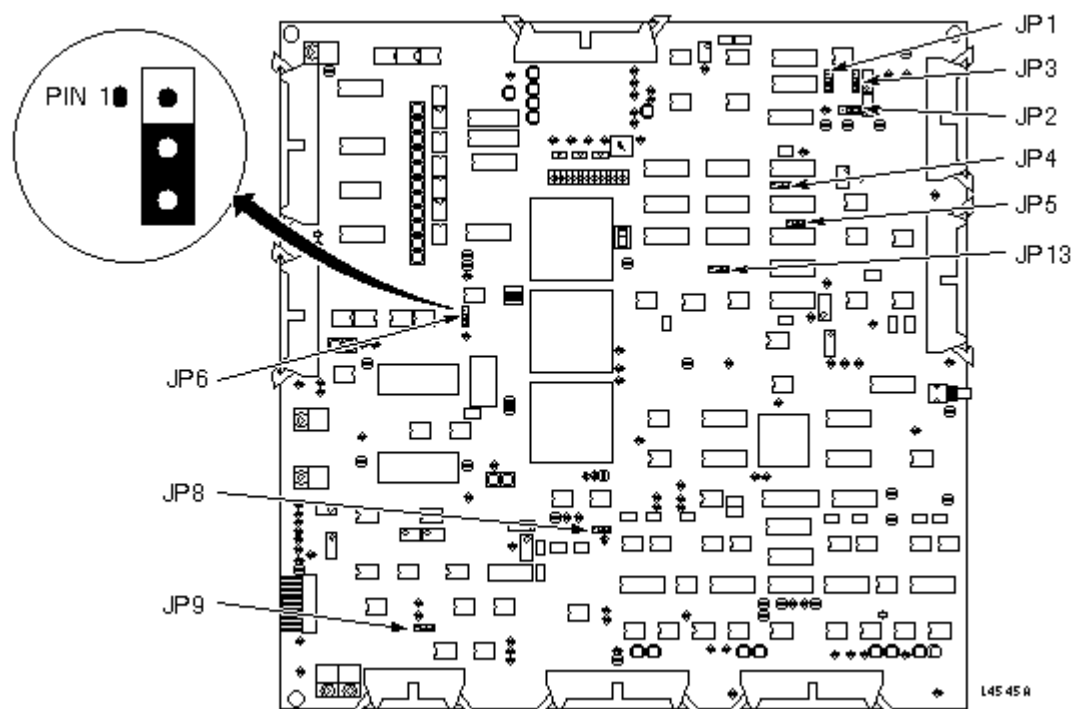


GRAM CONTROL BOARD SWITCH LOCATIONS
ILLUSTRATION L4542A

- b. See Table 2 and Illustration L4545A for proper jumper settings for the GRAM Control Board.

TABLE 2
GRAM CONTROL BOARD JUMPER SETTINGS

Jumper	Position	Setting
JP1	2 & 3	
JP2	2 & 3	
JP3	2 & 3	
JP4	2 & 3	(V/D) Voltage or Digital to Analog Converter – DAC selected
JP5	2 & 3	(AN/DIG) Analog or Digital – Digital selected
JP6	2 & 3	(DIS/EN) Disable or Enable – Enable selected
JP8	2 & 3	(TST/NRM) Test or Normal – Normal selected
JP9	2 & 3	Sets up one current sensor. Located <u>under</u> the Tuning Board
JP13	2 & 3	(AN/DIG) Charge mode

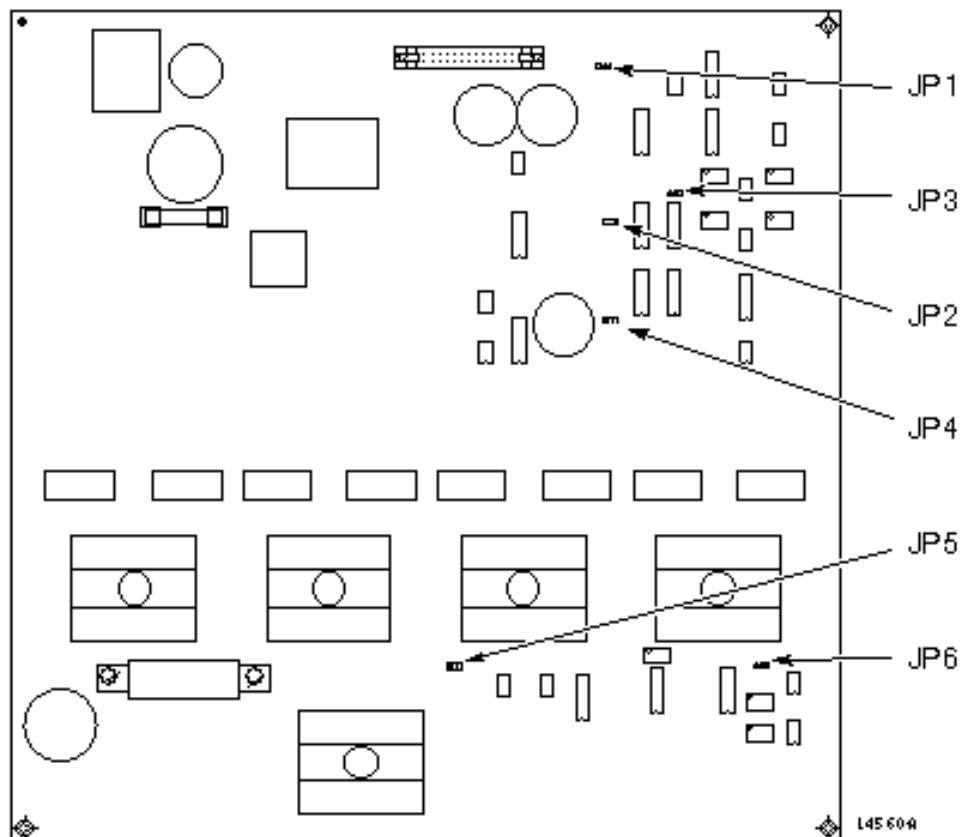


GRAM CONTROL BOARD JUMPER LOCATIONS
ILLUSTRATION L4545A

c. See Table 3 and Illustration L4560A for proper jumper settings for the GRAM Power Supply Board.

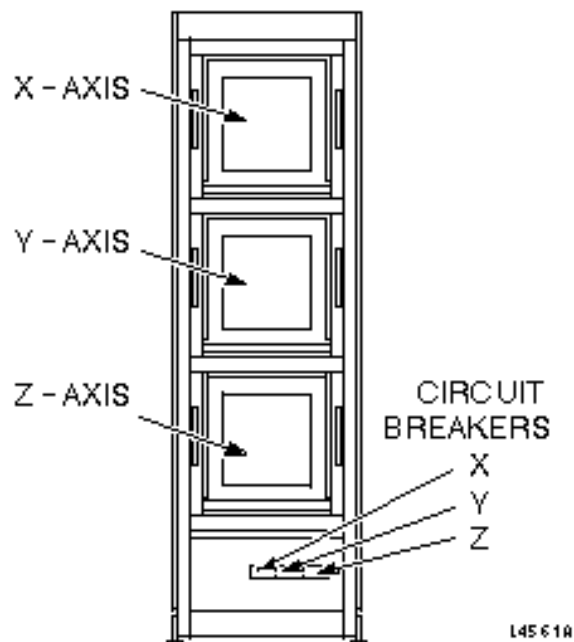
TABLE 3
GRAM POWER SUPPLY BOARD JUMPER SETTINGS

Name	Position Rev B	Position Rev C	Setting
JP1	2 & 3	2 & 3	Triangle enable
JP2	1 & 2	2 & 3	Connects LGND to AGND
JP3	1 & 2	2 & 3	High voltage enable
JP4	2 & 3	2 & 3	Under voltage enable
JP5	1 & 2	2 & 3	High voltage enable
JP6	1 & 2	2 & 3	NORM – Normal mode



GRAM POWER SUPPLY BOARD JUMPER LOCATIONS
ILLUSTRATION L4560A

4. Turn power to the GRAM cabinet off using the circuit breakers located at the lower front of the GRAM cabinet for all three GRAM modules (see Illustration L4561A).

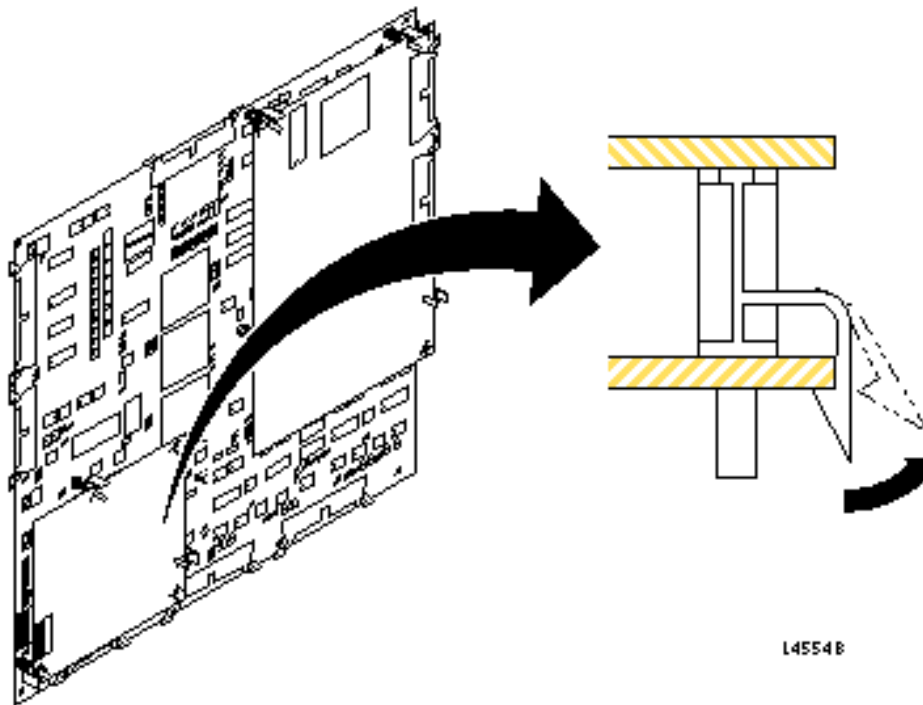


GRAM CABINET, FRONT VIEW
ILLUSTRATION L4561A



Equipment damage possibility. Do not pull the circuit board off. The nylon standoffs are quite brittle and will break. Push the top of the standoff away from the circuit board one by one. If they do not push off of the circuit board at first, work on the other standoffs iteratively.

5. Carefully push back the white nylon circuit board standoffs on the GRAM Analog Tuning board (see Illustration L4554B). Do not disconnect the GRAM Tuning Board at this time.



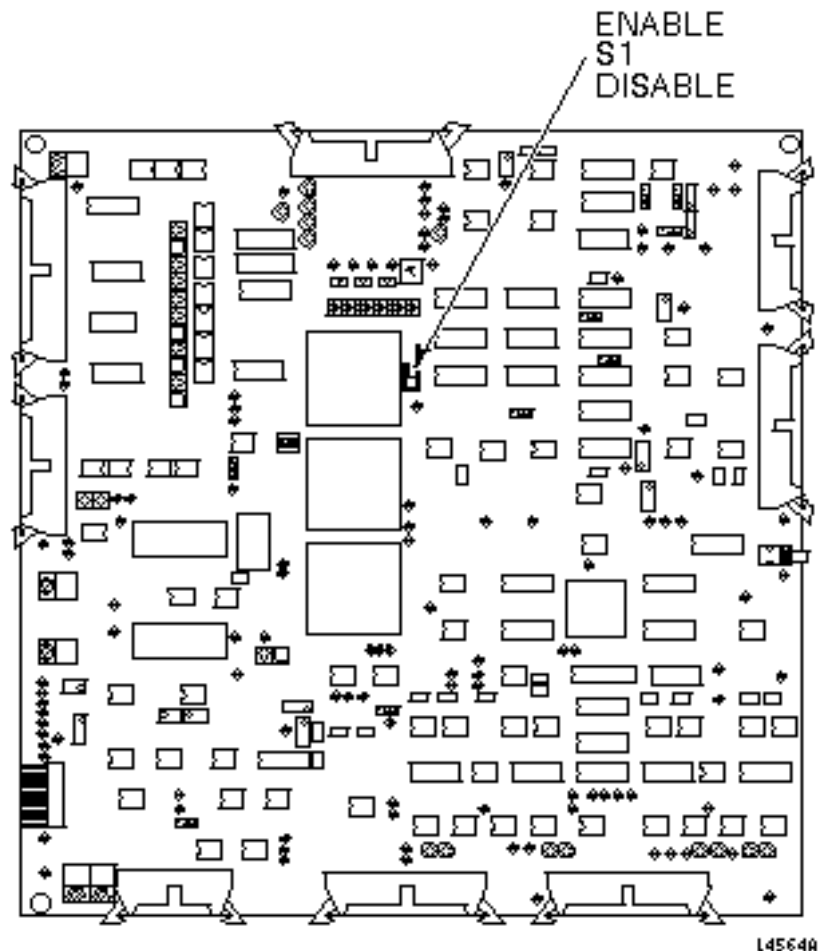
WHITE NYLON CIRCUIT BOARD STAND-OFFS
ILLUSTRATION L4554B

6. Use the nylon standoffs as a circuit board holder by turning the GRAM Analog Tuning Board perpendicular to the GRAM Control Board. Push the Tuning Board in between the clip and the post of one nylon standoff. This should hold the GRAM Analog Tuning Board in place while you perform the other adjustments.
7. If all three axes are to be calibrated, repeat steps 6 through 12 for each axis. If one or two axes are to be calibrated, perform steps 6 through 12 accordingly.
8. Turn on power to the GRAM cabinet by using the circuit breakers located at the lower-front of the GRAM cabinet for all three GRAM modules.

2- GRAM DC OFFSET ADJUSTMENT

Note - DC Offset adjustment - GRAM DC Offset adjustment must be performed first, or the return signal may not show up due to a magnitude offset caused by a DC offset on the output of the GRAM.

1. At the GRAM Control Board, verify that switch S1 is in the up/enable position (see Illustration L4564A).



S1 LOCATION
ILLUSTRATION L4564A

2. Landmark on the center line of the phantom.
3. At keypad on front magnet enclosure, press LANDMARK and Move to Scan.

At operator work space, prepare system to scan using the scan parameters shown in the "Service Protocols" procedure located under [Software Utilities] menu on the service methods CD-ROM, or for alternate proprietary procedure, use Table 4.

TABLE 4
GRAFIDY PROTOCOL

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

1. Click on **[New Pt]**, and enter

Id: **geservice**

Name: **grafidy**

Weight (Lb): **111**

Set Patient Protocols to **Service**.

2. Depending on your Release,

8.1: Click on **Other** to go to protocol list, then select **Grafidy**. Select series **1**.

8.2: In the Protocol field, type **o.5.1** (o=Other, 1=series number).

3. **[Save Series]**, then **[Prepare to Scan]**.

If you need help, see the User Interface Tutorial.

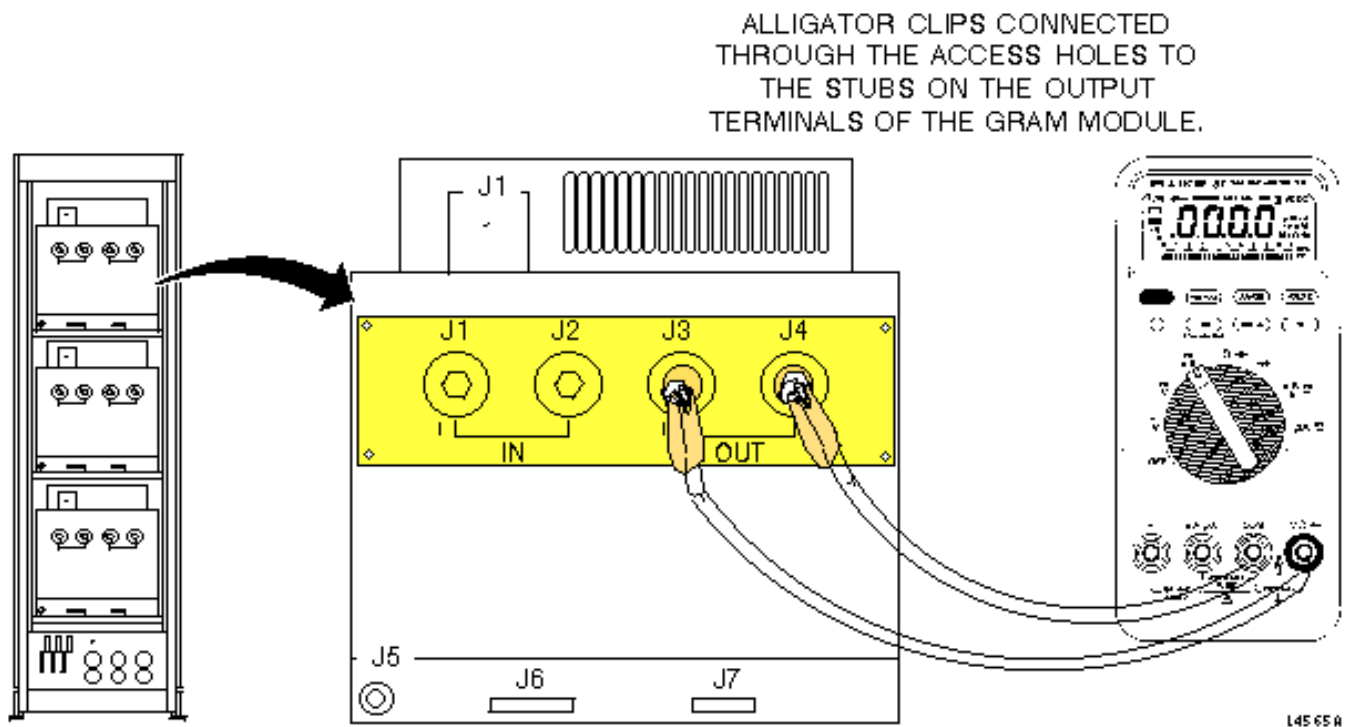
5. Go to manual prescan. Record grad shim values for x, y, and z axes. If values are not set to 0, set them by moving the slide bar to 0 at this time.
6. Select **[Done]** to exit manual prescan. This is important so that the scanner is ready, but not scanning.

Note - Resetting values - It is necessary to set the Grad Shim values to zero in order to ensure that there are no additional offsets in the circuit. This guarantees an accurate calibration.



FATAL ELECTRIC SHOCK HAZARD!! GRAM MODULES GENERATE FATAL ELECTRICAL CURRENTS. SCANNING OR PRESCANNING DURING THIS MEASUREMENT MAY CAUSE A FAILURE TO THE GRAM, AND EXPLOSION OF THE VOLTMETER. DO NOT SCAN OR PRESCAN DURING THIS MEASUREMENT AND ADJUSTMENT.

7. At the GRAM Control Board, prepare to perform the GRAM DC Current Offset Adjustment to null any offset currents to zero.
8. Using voltmeter leads with alligator clips, connect a voltmeter at the rear of the GRAM chassis across J3 (+OUT) and J4 (-OUT). See Illustration L4565A.



GRAM CHASSIS: REAR VIEW WITH VOLTMETER ACROSS THE OUTPUT
ILLUSTRATION L4565A

Note

Longer leads - Using longer leads for the voltmeter allows you to view the voltmeter as you perform the adjustment.

9. On the GRAM Control Board, locate and adjust the R151 (Offset Null) pot, so that the voltmeter reads 0 Vdc ± 0.010 Vdc. R151 is located under the GRAM Tuning Board (see Illustration L4564B in Section 2).

Note

DC offset adjustment - GRAM DC Offset Adjustment must be done with the system in Ready. If the fans in the 8645 Gradient Cabinet slow down while you perform the GRAM DC Offset Adjustment, GAP has put the Power Modules in Standby. Touch the Download soft key on the Scan Operations page, and the GAP will command the Power Modules to Ready.

10. If all three axes are to be calibrated, repeat steps 7 through 9 for each axis. If one or two axes are to be calibrated, repeat steps 7 through 9 accordingly.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 15, 1998	J. Saperstein	Initial conversion from Toolbook to Word formats.